Markov Decision Processes [1]

Salvador Ruiz Correa

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[1] Derived from Sutton & Barto conventions and the graphical model perspective.

This handout develops the Bellman expectation and optimality equations for Markov Decision Processes using the language of directed graphical models. By treating the MDP as a dynamic Bayesian network, the recurrence relations emerge naturally from d-separation and marginalization rather than from heuristic manipulation of expectations. We also make explicit the construction of the trajectory probability measure and the sense in which the value function is a conditional expectation under that measure. It is not common to use graphical model tools to present MDPs. We take into consideration Pearl's interventional calculus ideas, which justify our presentation. The transition probability $p(S_{t+1} \mid S_t, A_t)$ is the intervention distribution $p(S_{t+1} \mid S_t, \mathsf{do}(A_t))$

THE CENTRAL OBJECT of study in reinforcement learning is the *Markov Decision Process* (MDP). In the Sutton & Barto formalism, an MDP is a tuple $(\mathcal{S}, \mathcal{A}, p, \gamma)$ where $p(s', r \mid s, a)$ encodes the environment dynamics; $\pi(a \mid s) := p(a \mid s)$ encodes an agent policy .[2] Rather than treating this tuple as an algebraic abstraction, we view it as a *factorization* of a probability distribution over trajectories.

[2] Sutton, R. S. and Barto, A. G., *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

The MDP as a Dynamic Bayesian Network

At each time step t we have random variables $S_t \in \mathcal{S}$, $A_t \in \mathcal{A}$, $R_{t+1} \in \mathbb{R}$, and $S_{t+1} \in \mathcal{S}$ (Figure 1). The joint distribution over an infinite trajectory factorizes as

$$\begin{aligned} P_\pi(\tau) &= P(S_0) \prod_{t=0}^{\infty} \pi(A_t \mid S_t)\, p(R_{t+1} \mid S_t, A_t, S_{t+1})\, p(S_{t+1} \mid S_t, A_t) && (1) \\ &= P(S_0) \prod_{t=0}^{\infty} \pi(A_t \mid S_t)\, p(S_{t+1}, R_{t+1} \mid S_t, A_t), && (2) \end{aligned}$$

where $\tau = (S_0, A_0, R_1, S_1, A_1, S_2, A_2, R_2, \ldots)$, $p(s', r \mid s, a)$ encodes the environment dynamics, and $\pi(a \mid s)$ encodes an agent policy.

Equation 2 defines a directed acyclic graph (DAG) unrolled over time. The local Markov property encoded in this DAG is the statement that the future is conditionally independent of the past given the present state–action pair:

$$S_{t+1}, R_{t+1} \perp\!\!\!\perp S_{t-1}, A_{t-1}, \ldots, S_0 \mid S_t, A_t.$$

This is the *graphical model rendering* of the Markov property.

In plate notation one would draw a repeating time-slice; here we unroll the plate to make the d-separation arguments explicit.

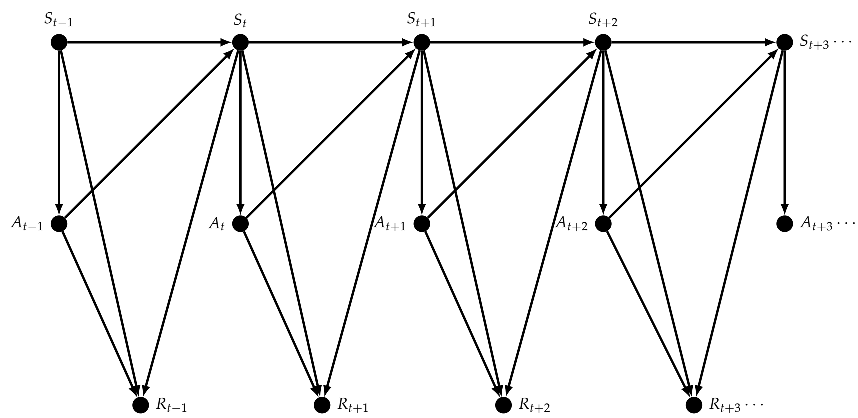

Figure 1: A section of the graphical model of an MDP.

The Return and Value Functions

Define the *discounted return* from time t as the random variable

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}, \qquad 0 \leq \gamma < 1.$$

The state-value and action-value functions are conditional expectations of this return:

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a],$$

that is,

$$v_\pi(s) = \mathbb{E}_\pi[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s], \qquad q_\pi(s,a) = \mathbb{E}_\pi[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a].$$

OUR GOAL is to show that these expectations satisfy linear recurrence relations—the Bellman equations—and to make explicit *why* the graphical model structure forces this recurrence.

The Trajectory Distribution and the Joint Law of (S_t, G_t)

Before proving the recurrence, we must be precise about the probability space on which these expectations are defined.

The base measure. The MDP defines a probability measure P_π on the space of infinite trajectories

$$\Omega = (\mathcal{S} \times \mathcal{A} \times \mathbb{R})^{\mathbb{N}}$$

via the factorization in equation 2. This is the set of all infinite sequences (countably infinite) of triples S_t, A_t, R_t indexed by the natural numbers. Every random variable we care about, S_t, A_t, R_{t+1}, and therefore G_t, lives on this space.

Notice that the cardinality $|\mathcal{S}|$ can be finite (grid worlds, chessboard positions, card games), countable infinite (queuing systems, inventory levels, a counter that increments forever), or uncountable infinite (robotics: joint angles, positions, velocities; autonomous driving: continuous coordinates, any physical system with continuous states).

G_t is a derived random variable. G_t is *not* a node in the original factorization. It is a measurable function of the trajectory:

$$G_t : \Omega \to \mathbb{R}, \qquad G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}(\tau).$$

Because $|\gamma| < 1$ and rewards are bounded, this series converges absolutely P_π-almost surely, so G_t is a well-defined random variable.

Conditional expectation as regression. From the graphical model perspective, $v_\pi(s)$ is the *conditional expectation* (regression function) of the derived variable G_t on the observed variable S_t:

$$v_\pi(s) = \mathbb{E}_{P_\pi}[G_t \mid S_t = s] = \int g \, dP_\pi(G_t = g \mid S_t = s).$$

The Bellman proof below uses the tower property to write this as

$$\mathbb{E}_{P_\pi}[G_t \mid S_t = s] = \sum_a \pi(a \mid s) \, \mathbb{E}_{P_\pi}[G_t \mid S_t = s, A_t = a],$$

which is just marginalizing out the action variable A_t from the joint distribution of (S_t, A_t, G_t).

Proof of the Bellman Expectation Equation

Theorem 1 (Bellman Expectation Equation) *For any policy π, the action-value function satisfies*

$$q_\pi(s,a) = \sum_{s',r} p(s', r \mid s, a) \Big[r + \gamma \, v_\pi(s') \Big],$$

and the state-value function satisfies

$$v_\pi(s) = \sum_a \pi(a \mid s) \, q_\pi(s,a).$$

Proof 1 *We proceed by graphical model marginalization.*

Step 1. Conditioning on the immediate action. *We begin with the definition of the state-value function: it is the conditional expectation of the return random variable G_t given that the process is in state s at time t:*

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s].$$

In the unrolled directed graphical model, S_t is a parent of A_t via the policy edge $S_t \rightarrow A_t$, but A_t is not *determined once we condition on $S_t = s$. Rather, A_t remains a random variable governed by the policy, with conditional distribution $P(A_t = a \mid S_t = s) = \pi(a \mid s)$. Because the return G_t depends on the entire future trajectory, and the first stochastic branching point of that future (given the evidence $S_t = s$) is the choice of action, we cannot compute the expectation of G_t without first accounting for the uncertainty in A_t.*

To make this precise we invoke the law of total expectation *(also called the tower property or marginalization over the action variable). For any three random variables X, Y, and Z with Y discrete, the tower property states*

$$\mathbb{E}[X \mid Z] = \sum_y \mathbb{E}[X \mid Z, Y = y]\, P(Y = y \mid Z).$$

Applying this with $X = G_t$, $Z = S_t$, and $Y = A_t$ yields

$$\mathbb{E}_\pi[G_t \mid S_t = s] = \sum_{a \in \mathcal{A}} \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]\, P(A_t = a \mid S_t = s).$$

The first factor in each term is exactly the action-value function $q_\pi(s, a)$ by definition. The second factor is the policy probability $\pi(a \mid s)$, because the joint distribution over trajectories factorizes as in equation (2)*; conditioning on $S_t = s$ isolates the factor $\pi(A_t \mid S_t = s)$, so the conditional probability of $A_t = a$ is precisely $\pi(a \mid s)$. Therefore*

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s)\, q_\pi(s, a). \tag{1}$$

Graphical-model interpretation. *This step is nothing more than* variable elimination *in the DAG. We are computing the expected value of a downstream function (G_t) given evidence at an ancestor node ($S_t = s$). The immediate child A_t lies on every directed path from S_t to the future reward variables, so to propagate the evidence forward we must sum A_t out of the factor graph, weighting each instantiation by its local conditional probability $\pi(a \mid s)$. The result is a message from the action node to the state node: a weighted average of the action-specific expectations $q_\pi(s, a)$.*

Measure-theoretic remark. *On the probability space induced by the policy π, the conditional expectation $\mathbb{E}_\pi[G_t \mid S_t]$ is a random variable measurable with respect to the σ-algebra generated by S_t. The tower property $\mathbb{E}[\mathbb{E}[G_t \mid S_t, A_t] \mid S_t] = \mathbb{E}[G_t \mid S_t]$ is exact; because A_t takes countably many values, the inner conditional expectation disintegrates into the discrete sum shown above. In the continuous-action case the sum becomes an integral $\int_{\mathcal{A}} \pi(a \mid s)\, q_\pi(s, a)\, da$, but the graphical-model reasoning—marginalizing out the intermediate action node—remains identical.*

Policy matrix viewpoint. *If one arranges the state values into a vector $\mathbf{v}_\pi \in \mathbb{R}^{|\mathcal{S}|}$ and the action values into a vector $\mathbf{q}_\pi \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, this equation becomes the linear relation $\mathbf{v}_\pi = \Pi\, \mathbf{q}_\pi$, where Π is the $|\mathcal{S}| \times |\mathcal{S}||\mathcal{A}|$ matrix* *with entries $\Pi_{(s),(s,a)} = \pi(a \mid s)$. Thus Step 1 is the linear projection of the action-value vector onto the state-value vector induced by the policy. The linearity arises because marginalization is a linear operator on the space of probability distributions over the graphical model.*

Step 2. Recursive decomposition of the return. *The return satisfies the identity of random variables*

$$G_t = R_{t+1} + \gamma G_{t+1}.$$

Substituting this into the definition of q_π and using linearity of expectation gives

$$q_\pi(s,a) = \mathbb{E}_\pi[R_{t+1} \mid S_t = s, A_t = a] + \gamma\, \mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a]. \quad (2)$$

Step 3. Marginalizing over the next state–reward pair. *The first term in (2) is direct from the dynamics:*

$$\mathbb{E}_\pi[R_{t+1} \mid S_t = s, A_t = a] = \sum_{s',r} r\, p(s',r \mid s,a).$$

For the second term, we marginalize over S_{t+1} and R_{t+1} using the trajectory factorization (2):

$$\mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a] = \sum_{s',r} p(s',r \mid s,a)\, \mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s', R_{t+1} = r]. \quad (3)$$

Step 4. d-separation (the Markov property). *In the unrolled DAG, G_{t+1} is a function of the sub-graph rooted at S_{t+1}. Every path from the past (S_t, A_t) to G_{t+1} is blocked by conditioning on S_{t+1}. Hence*

$$G_{t+1} \perp\!\!\!\perp S_t, A_t, R_{t+1} \mid S_{t+1}.$$

The conditional expectation in (3) therefore collapses to

$$\mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s'] = v_\pi(s').$$

Step 5. Assembly. *Substituting back into (2) yields*

$$q_\pi(s,a) = \sum_{s',r} p(s',r \mid s,a)\big[\, r + \gamma\, v_\pi(s')\,\big],$$

and combining with (1) gives the full recurrence for v_π.

The critical step is Step 4: the recurrence arises not from algebraic trickery but from the *conditional independence structure* of the DAG.

This is why the Bellman equations are *theorems* of the graphical model, not merely definitions. The self-similarity of the unrolled DAG makes the future subgraph isomorphic to the whole, forcing the recurrence.

Proof of the Bellman Optimality Equation

Theorem 2 (Bellman Optimality Equation) *The optimal state-value function $v_*(s) = \max_\pi v_\pi(s)$ satisfies*

$$v_*(s) = \max_{a \in \mathcal{A}} \sum_{s',r} p(s', r \mid s, a) \Big[r + \gamma \, v_*(s') \Big],$$

and the optimal action-value function satisfies

$$q_*(s, a) = \sum_{s',r} p(s', r \mid s, a) \Big[r + \gamma \max_{a'} q_*(s', a') \Big].$$

Proof 2 *The environment dynamics $p(s', r \mid s, a)$ are fixed structural elements of the graphical model; only the policy π is subject to optimization.*

For a fixed action a, the best achievable future value from the next state is $v_*(s')$. By the same d-separation argument as above, the future subgraph from S_{t+1} is independent of the choice of A_t once S_{t+1} is known. Hence

$$q_*(s, a) = \sum_{s',r} p(s', r \mid s, a) \big[r + \gamma \, v_*(s') \big].$$

Because we may choose the action at each state independently, optimality requires selecting the action that maximizes the action-value:

$$v_*(s) = \max_{a \in \mathcal{A}} q_*(s, a).$$

Substituting the expression for q_* yields the stated recurrence. The q-form follows by replacing $v_*(s')$ with $\max_{a'} q_*(s', a')$.

Summary: Graphical Model Operations

Graphical Model Operation	RL Concept	Equation Step
DAG factorization	MDP definition	Eq. (2)
d-separation on unrolled graph	Markov property	Step 4 of Theorem 1
Law of total expectation (marginalize A_t)	Policy evaluation	$v_\pi(s) = \sum_a \pi(a \mid s) q_\pi(s, a)$
Law of total expectation (marginalize S_{t+1}, R_{t+1})	Expectation over transitions	Step 3 of Theorem 1
Self-similarity of time-unrolled DAG	Recursion $G_t = R_{t+1} + \gamma G_{t+1}$	Step 2 of Theorem 1
Maximization over action nodes	Bellman optimality	Theorem 2

Object	What it is	How to get it
$P_\pi(\tau)$	Joint over full trajectory	Product of local factors (the DAG)
$P_\pi(S_t = s)$	Marginal state distribution	Sum $P_\pi(\tau)$ over all τ with $S_t = s$
$P_\pi(S_t = s, G_t \in B)$	Joint of state and return	Sum $P_\pi(\tau)$ over all τ with $S_t = s$ and $G_t \in B$
$v_\pi(s)$	Conditional expectation	Use Bellman recurrence (avoids explicit marginalization)

In summary, the Bellman equations are not merely definitions of the value functions; they are *consequences* of the conditional independence structure of the MDP viewed as a dynamic Bayesian network.

The recurrence emerges from marginalizing over the immediate variables and exploiting the Markov property via d-separation. The underlying trajectory measure P_π gives rigorous meaning to every expectation, while the graphical model provides the algorithmic structure that makes the recurrences computable.

The Law of Total Expectation

The Bellman proof in the preceding section relies on the identity

$$\mathbb{E}[G_t \mid S_t = s] = \sum_{a \in \mathcal{A}} \mathbb{E}[G_t \mid S_t = s, A_t = a]\, P(A_t = a \mid S_t = s).$$

This is an instance of the *law of total expectation* (also called the *tower property*). We prove it twice: first in the discrete setting needed for the MDP, and then in full generality.

The Discrete Case

Let X, Y, and Z be random variables defined on a common probability space $(\Omega, \mathcal{F}, P)$, with Y taking values in a countable set $\mathcal{Y}$. Assume $\mathbb{E}|X| < \infty$.

Theorem 3 (Law of Total Expectation — Discrete Conditioning) *For any z with $P(Z = z) > 0$,*

$$\mathbb{E}[X \mid Z = z] = \sum_{y \in \mathcal{Y}} \mathbb{E}[X \mid Y = y, Z = z]\, P(Y = y \mid Z = z).$$

Proof 3 *We work from the definition of conditional expectation for events. Recall that for any two events A and B with $P(B) > 0$, the conditional probability is $P(A \mid B) = P(A \cap B)/P(B)$. For a discrete random variable Y, the conditional probability mass function is*

$$P(Y = y \mid Z = z) = \frac{P(Y = y, Z = z)}{P(Z = z)}.$$

Step 1. Expand the left-hand side by definition. *By the definition of conditional expectation given an event,*

$$\mathbb{E}[X \mid Z = z] = \int_\Omega X(\omega)\, dP(\omega \mid Z = z) = \frac{1}{P(Z = z)} \int_{\{Z=z\}} X\, dP. \tag{4}$$

Step 2. Partition the sample space. *Because Y is discrete, the event $\{Z = z\}$ partitions into the countable disjoint union*

$$\{Z = z\} = \bigsqcup_{y \in \mathcal{Y}} \{Y = y, Z = z\}.$$

Therefore, by countable additivity of the integral,

$$\int_{\{Z=z\}} X\, dP = \sum_{y \in \mathcal{Y}} \int_{\{Y=y, Z=z\}} X\, dP. \tag{5}$$

Step 3. Recognize conditional expectations. *For each y with $P(Y = y, Z = z) > 0$, the conditional expectation $\mathbb{E}[X \mid Y = y, Z = z]$ is defined as*

$$\mathbb{E}[X \mid Y = y, Z = z] = \frac{1}{P(Y = y, Z = z)} \int_{\{Y=y, Z=z\}} X \, dP.$$

Multiplying both sides by $P(Y = y, Z = z)$ and substituting into (5) yields

$$\int_{\{Z=z\}} X \, dP = \sum_{y \in \mathcal{Y}} \mathbb{E}[X \mid Y = y, Z = z] \, P(Y = y, Z = z). \tag{6}$$

Step 4. Divide by $P(Z = z)$. *Substituting (6) into (4) and factoring $P(Z = z)$ into the sum gives*

$$\mathbb{E}[X \mid Z = z] = \sum_{y \in \mathcal{Y}} \mathbb{E}[X \mid Y = y, Z = z] \, \frac{P(Y = y, Z = z)}{P(Z = z)}.$$

The ratio is exactly $P(Y = y \mid Z = z)$, completing the proof.

Graphical model reading. In factor-graph language, this proof is the operation of *sum-marginalization*: we partition the configuration space by the values of Y, compute the local expectation in each slice, and reweight by the marginal probability of that slice.[3]

[3] This is why variable elimination in a Bayesian network is algorithmically identical to the tower property.

The General Case

For continuous or mixed random variables, the sum becomes an integral and the proof requires the measure-theoretic definition of conditional expectation. We state the result in the language of random variables and σ-algebras.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let X be integrable. Let Y and Z be random variables. Denote by $\sigma(Z)$ the σ-algebra generated by Z, and by $\sigma(Y, Z)$ the σ-algebra generated by the pair (Y, Z). Note that $\sigma(Z) \subseteq \sigma(Y, Z)$.

Theorem 4 (Tower Property)

$$\mathbb{E}\big[\, \mathbb{E}[X \mid Y, Z] \mid Z \big] = \mathbb{E}[X \mid Z] \quad P\text{-a.s.}$$

Proof 4 *We use the defining characterization of conditional expectation. Recall that a random variable U equals $\mathbb{E}[X \mid \mathcal{G}]$ almost surely if and only if (i) U is $\mathcal{G}$-measurable, and (ii) $\int_A U \, dP = \int_A X \, dP$ for all $A \in \mathcal{G}$.*

Set $U = \mathbb{E}[X \mid Z]$ and $V = \mathbb{E}[X \mid Y, Z]$. We must show that $\mathbb{E}[V \mid Z] = U$.

Measurability. *Both U and $\mathbb{E}[V \mid Z]$ are $\sigma(Z)$-measurable by definition.*

Integral identity. *Let $A \in \sigma(Z)$. Because $\sigma(Z) \subseteq \sigma(Y, Z)$, we also have $A \in \sigma(Y, Z)$. Applying the defining property to $V = \mathbb{E}[X \mid Y, Z]$ on the set $A \in \sigma(Y, Z)$ gives*

$$\int_A V \, dP = \int_A X \, dP. \tag{7}$$

Now apply the defining property to $\mathbb{E}[V \mid Z]$ on the set $A \in \sigma(Z)$:

$$\int_A \mathbb{E}[V \mid Z] \, dP = \int_A V \, dP. \tag{8}$$

Combining (7) and (8) yields

$$\int_A \mathbb{E}[V \mid Z] \, dP = \int_A X \, dP \qquad \text{for all } A \in \sigma(Z). \tag{9}$$

But $U = \mathbb{E}[X \mid Z]$ is the unique *(a.s.) $\sigma(Z)$-measurable random variable satisfying (9). Hence $\mathbb{E}[V \mid Z] = U$ almost surely.*

From σ-algebras to states and actions. To recover the discrete formula used in the MDP, take $Z = S_t$ and $Y = A_t$. Because A_t is discrete, the conditional expectation $\mathbb{E}[G_t \mid S_t]$ is a random variable that can be written explicitly as

$$\mathbb{E}[G_t \mid S_t = s] = \sum_a \mathbb{E}[G_t \mid S_t = s, A_t = a] \, P(A_t = a \mid S_t = s),$$

which is exactly the formula invoked in Step 1 of the Bellman proof.

Remark on densities. If (Y, Z) has a joint density $f_{Y,Z}(y, z)$ and X is a function of the underlying trajectory, the tower property becomes

$$\mathbb{E}[X \mid Z = z] = \int_{\mathcal{Y}} \mathbb{E}[X \mid Y = y, Z = z] \, f_{Y|Z}(y \mid z) \, dy,$$

where $f_{Y|Z}(y \mid z) = f_{Y,Z}(y, z) / f_Z(z)$. The proof is identical in structure, replacing sums by integrals and probability mass functions by densities. The graphical model interpretation—marginalizing out an intermediate variable—remains unchanged.

Measure-Theoretic Proof

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let X be an integrable random variable. Let $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \mathcal{F}$ be nested sub-σ-algebras.

Theorem 5 (Tower Property)

$$\mathbb{E}\big[\, \mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1 \big] = \mathbb{E}[X \mid \mathcal{G}_1] \quad P\text{-a.s.}$$

Proof 5 *Recall the defining characterization: a random variable U equals $\mathbb{E}[X \mid \mathcal{G}]$ a.s. iff (i) U is $\mathcal{G}$-measurable and (ii) $\int_A U \, dP = \int_A X \, dP$ for all $A \in \mathcal{G}$.*

Set $U = \mathbb{E}[X \mid \mathcal{G}_1]$ and $V = \mathbb{E}[X \mid \mathcal{G}_2]$. We show $\mathbb{E}[V \mid \mathcal{G}_1] = U$.

Measurability. *Both U and $\mathbb{E}[V \mid \mathcal{G}_1]$ are $\mathcal{G}_1$-measurable by definition.*

Integral identity. *Let $A \in \mathcal{G}_1$. Since $\mathcal{G}_1 \subseteq \mathcal{G}_2$, also $A \in \mathcal{G}_2$. By the defining property of V on the set $A \in \mathcal{G}_2$:*

$$\int_A V\,dP = \int_A X\,dP. \tag{10}$$

By the defining property of $\mathbb{E}[V \mid \mathcal{G}_1]$ on the set $A \in \mathcal{G}_1$:

$$\int_A \mathbb{E}[V \mid \mathcal{G}_1]\,dP = \int_A V\,dP. \tag{11}$$

Combining (10) and (11):

$$\int_A \mathbb{E}[V \mid \mathcal{G}_1]\,dP = \int_A X\,dP \qquad \text{for all } A \in \mathcal{G}_1. \tag{12}$$

But $U = \mathbb{E}[X \mid \mathcal{G}_1]$ is the unique a.s. $\mathcal{G}_1$-measurable random variable satisfying (12). Hence $\mathbb{E}[V \mid \mathcal{G}_1] = U$ a.s.

To recover the formula used in the MDP, set $\mathcal{G}_1 = \sigma(S_t)$ and $\mathcal{G}_2 = \sigma(S_t, A_t)$. Because A_t is discrete, evaluating the conditional expectations at the atoms yields

$$\mathbb{E}[G_t \mid S_t = s] = \sum_a \mathbb{E}[G_t \mid S_t = s, A_t = a]\, P(A_t = a \mid S_t = s).$$

Graphical Model Marginalization Proof

We now prove the same identity by interpreting the trajectory distribution as a factor graph and performing variable elimination.[4]

[4] This proof is algorithmic: it shows *why* the tower property is true by exhibiting the sequence of local operations on the graph.

The factor graph. The trajectory distribution (2) factorizes as

$$P_\pi(\tau) = \phi_0(S_0) \prod_{t=0}^{\infty} \phi_t^\pi(S_t, A_t)\, \psi_t(S_t, A_t, S_{t+1}, R_{t+1}),$$

where $\phi_0(s) = P(S_0 = s)$, $\phi_t^\pi(s,a) = \pi(a \mid s)$, and $\psi_t(s,a,s',r) = p(s', r \mid s, a)$. In the factor graph, each factor node is connected to its argument variables.

Query. We wish to compute $v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$. By definition of expectation, this is

$$v_\pi(s) = \int g\, dP_\pi(G_t = g \mid S_t = s) = \frac{\sum_\tau G_t(\tau)\, P_\pi(\tau)\, \mathbf{1}_{\{S_t(\tau)=s\}}}{\sum_\tau P_\pi(\tau)\, \mathbf{1}_{\{S_t(\tau)=s\}}}.$$

Eliminate A_t. Focus on the subgraph containing S_t, A_t, and all downstream variables. Because the graph is directed and acyclic, the joint over this subgraph factorizes as

$$P_\pi(S_t = s, A_t = a, \text{future}) = \pi(a \mid s)\, p(\text{future} \mid s, a).$$

To marginalize A_t out of the numerator, we sum over $a \in \mathcal{A}$:

$$\sum_a \pi(a \mid s) \sum_{\text{future}} G_t(\tau)\, p(\text{future} \mid s, a).$$

The inner sum is precisely $\mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$ times the normalizing constant of the future (which is 1 because p is a probability distribution). Hence the numerator becomes

$$\sum_a \pi(a \mid s)\, \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \;\times\; P_\pi(S_t = s).$$

The denominator is simply $P_\pi(S_t = s)$. Cancelling gives

$$v_\pi(s) = \sum_a \pi(a \mid s)\, \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a],$$

which is exactly the tower property for this query.

Why this is the same proof. In the measure-theoretic proof, equation (12) asserts that the marginal expectation over a coarser σ-algebra equals the integral of the conditional expectation. In the factor graph proof, we perform the same operation explicitly: we partition the configuration space by the values of A_t (the coarse σ-algebra $\sigma(S_t)$ is the quotient that forgets A_t), compute the local integral in each fiber (the conditional expectation), and reweight by the marginal probability of the fiber (the factor $\pi(a \mid s)$). The two proofs are identical; the factor graph simply makes the *local* nature of the marginalization visible.

In a junction tree algorithm, the message passed from the (S_t, A_t) clique to the S_t clique is exactly the vector of action-value expectations $q_\pi(s, \cdot)$. The marginalization step is matrix-vector multiplication by the policy matrix Π.